



Al-Tahtheeb Educational Services
شركة التحذيب للخدمات التعليمية

DIGITAL TRANSFORMATION ROADMAP 2026

Smarter Schools, Empowered Futures.

مدارس أذكى – مستقبل أكثر إشراقاً

A comprehensive one-year digital transformation program designed for Al-Tahtheeb Educational Services — integrating AI, IoT, smart mobility, parent engagement, and administrative excellence across all campuses.

Artificial Intelligence

Parent Mobile App

Analytics Dashboard

IoT Smart Campus

Odoo 19 EMS

Bus Tracking

Nafath e-Contracts

Digital Payments

TRANSFORMATION PILLARS	48+	4	12
	INITIATIVES	QUARTERS	MONTHS
KSA VISION 2030 ALIGNED			

Why Digital Now?

لماذا التحول الرقمي الآن؟

Saudi Vision 2030 mandates educational excellence through digital innovation. Al-Talhtheeb's 2026 program positions every campus at the forefront of smart education.

100%

PAPERLESS REGISTRATION
تسجيل رقمي كامل

3 min

AVG. ENROLLMENT TIME
وقت تسجيل الطلاب

98%

PARENT APP ADOPTION
نسبة تفعيل تطبيق الوالدين

0

PAPER CONTRACTS
عقود ورقية

Live

BUS GPS TRACKING
تتبع الحافلات المباشر

AI

EARLY WARNING SYSTEM
نظام الإنذار المبكر

ZATCA

COMPLIANT INVOICING
فوترة متوافقة

EIGHT PILLARS OF TRANSFORMATION

Our Digital Pillars

ركائز التحول الرقمي الثماني

Each pillar targets a critical dimension of the educational ecosystem — from classroom technology to AI-driven analytics and smart campus infrastructure.



Student Management & EMS

إدارة الطلاب ونظام EMS

- Digital enrollment with Nafath e-contract authentication (Odoo 19 EMS)
- Yakeen & Muqeen API auto-fill for Saudi and non-Saudi students



Smart Classroom & EdTech

الفصل الذكي وتكنولوجيا التعليم

- Interactive smart boards in all classrooms (touchscreen, multi-input)
- E-Learning integration: Classera / Noor LMS platform / Classera



IoT Smart Campus Infrastructure

البنية التحتية الذكية للحم مدرسي

- Automated biometric attendance (facial recognition + fingerprint)



Smart Bus & Transportation

النقل المدرسي الذكي

- حضور بيومتري آلي (تعرف وجه +

- Automated installment invoicing & payment schedules
- Student academic progress tracking & promotion workflows

- Digital document vault per student (ID, contracts, reports)

- Financial enforcement: portal blocks & certificate withholding

Odoo 19 Nafath Yakeen Muqeen REST API

فوترة تعاقدية وجدول أقساط

تتبع التقدم الأكاديمي وسير ترقية الطالب

خزنة وثائق رقمية لكل طالب

تطبيق مالي، حظر البوابة وتجديد الشهادات

- STEAM Labs with Robotics and 3D printing stations

- Google Chromebook / iPad device program per grade

- Digital library with e-books, audiobooks, and research portals

- Gamified assessments & student engagement tools

Classera Google Workspace Kahoot 3D Print



AI & Learning Analytics

الداء الاصطناعي وتحليلات التعلم

- AI-powered early warning system for at-risk students

- Predictive enrollment analytics & capacity forecasting

- AI tutoring assistant: personalized learning recommendations

- Automated grading assistance for teachers (MCQ + short answer)

- Executive KPI dashboard: real-time enrollment, revenue, attendance

- Sentiment analysis on parent feedback and support tickets

Python / ML Odoo BI Power BI OpenAI API

Metabase



Parent Engagement & Communication

مشاركة الوالدين والتواصل

- Parent mobile app: invoices, grades, attendance, schedule, news

- WhatsApp Business bot: automated alerts & 2-way chat

- Live GPS bus tracking — know exactly when your child arrives

- Student gate entry/exit notifications (RFID + facial recognition)

- Online parent-teacher meeting scheduling

- Multilingual push notifications (Arabic + English)

WhatsApp API GPS/MQTT RFID Firebase

Odoo Portal



IoT Smart Campus Infrastructure

البنية التحتية الذكية للحم مدرسي

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Smart Bus & Transportation

النقل المدرسي الذكي

- حضور بيومتري آلي (تعرف وجه +

Flagship Innovations

أبرز الابتكارات التقنية

High-impact digital features that will directly transform the daily experience for students, parents, and staff.



Live Bus Tracking

تتبع الحافلة المدرسية مباشرة

Parents open the app and see their child's bus on a live map. Geo-fence alerts when the bus is 5 minutes away. RFID scan confirms boarding and alerts the parent instantly.

SAFETY - PARENTAL PEACE OF MIND



AI Early Warning System

نظام الإنذار المبكر بالذكاء الاصطناعي

Machine learning model analyzes attendance trends, grades, and behavioral data to flag at-risk students 4-6 weeks before academic failure — enabling early counselor intervention.

AI - STUDENT WELLBEING



Biometric Campus Entry

دخول الحرم بالتعرف البيومتري

Facial recognition at school gates logs student arrival in under 1 second. Parent is notified automatically. Unauthorized access triggers an instant security alert to the school admin.

SECURITY - IOT - AI VISION



Executive Analytics Dashboard

لوحة مؤشرات التحليلات التشغيلية

Real-time KPI dashboard for school directors and the CEO: live enrollment figures, revenue collection rates, attendance heatmaps, fee default alerts, and year-over-year comparisons.

BI - DECISION INTELLIGENCE



Nafath e-Contract Signing

توقيع العقد الإلكتروني عبر نفاذ

Parents sign school enrollment contracts digitally using their Nafath national identity. No paper. No branch visits. Legally binding and ZATCA-compliant. Registration cannot proceed without it.

LEGAL - COMPLIANCE - SAUDI NCA

HR رقمي: عقود، رواتب، إجازات، امتثال مقسم

نظام تذكرة داخلي للصيانة وطلبات المرافق

مشتريات رقمية: أوامر شراء وإدارة الموردين

إدارة الأصول: سجل معدات المدرسة والاستهلاك

CRM للتواصل مع الأولياء المحتملين وحملات القبول

إدارة الجودة، المؤشرات، التدقيق والتحسين المستمر

Muqem API

Odoo Purchase

Odoo HR

Odoo CRM

Q1-Q4

HR, Operations & Administrative Excellence

الموارد البشرية والعمليات الإدارية



بصقة

تكييف ذكي، تحكم IoT بالحرارة وجودة الهواء

إضاءة ذكية: LED بحساس الحركة لتوفير الطاقة

مراقبة CCTV بالذكاء الاصطناعي وتسجيل التسجيل

حساسات كشف تسرب المياه والحريق مع تنبيهات آلية

شاشات لإفادت رقمية، إعلانات فورية لكل حرم

MQTT

Raspberry Pi

AWS IoT

OpenCV

Zigbee

Q1-Q3

Financial Excellence & Compliance

التميز المالي والامتثال التنظيمي



فوترة إلكترونية متوافقة مع هيئة الزكاة (QR + XML)

Amazon Payment Services: Visa, Mada, SADAD, BNPL

دليل الحسابات السعودي + أتمتة ضريبة القيمة المضافة (15%)

تتبع الإيرادات المؤجلة IFRS 15 والإغلاق الشهري الآلي

تذكير آلي بالمفوعات للوفاء عبر SMS وWhatsApp

تحليلات مالية: تتبع المتأخرين وتوقع الإيرادات

Odoo 19

ZATCA Phase 2

Amazon FORT

IFRS 15

12-Month Roadmap

خارطة طريق 12 شهراً

Phased delivery across four quarters — each building on the previous to ensure stability and stakeholder readiness.

Every Parent, Always Connected.

كل ولي أمر – دائماً على تواصل

A seamless digital experience that keeps parents informed and engaged — 24/7, in Arabic and English.

2026 Digital Transformation — Quarterly Delivery Plan

خطة التسليم الفصلية لبرنامج التحول الرقمي 2026

INITIATIVE	Q1 - JAN-MAR	Q2 - APR-JUN	Q3 - JUL-SEP	Q4 - OCT-DEC
● Production Server + Odoo 19	DEPLOY	STABILIZE		
● EMS Core — Enrollment & Fees	CONFIG	GO-LIVE	OPTIMIZE	
● Nafath + Yakeen + Muqeem	DEV	TEST & LIVE		
● Saudi Accounting + ZATCA	SETUP	UAT	LIVE	
● Parent Mobile App + WhatsApp Bot		BUILD	BETA	FULL LAUNCH
● GPS Bus Tracking System		PILOT 1 SCHOOL	ALL SCHOOLS	ENHANCE
● IoT Campus (Biometric + CCTV)		INSTALL	ALL CAMPUSES	MONITORING
● IoT Smart Classrooms (HVAC + LED)			PILOT	ROLLOUT
● AI Early Warning System			TRAIN MODEL	LIVE
● Executive Analytics Dashboard		DESIGN	BUILD	LIVE
● Smart Classrooms + EdTech		PILOT	SCALE	FULL ROLLOUT
● HR + Procurement + CRM (Odoo)	CONFIG	GO-LIVE	OPTIMIZE	FULL USE
● Staff Training (All Roles)	KU TRAINING	STAFF TRAIN	REFRESH	ADVANCED

LEGEND: Active Work Not Started / Future



Parent Mobile App

تطبيق ولي الأمر

Q2

- View & pay invoices directly (Amazon / Mada / SADAD)
- Real-time attendance & grade notifications
- Download school reports & academic progress
- Book parent-teacher meetings online
- Submit leave/absence requests digitally
- Receive school announcements & events

React Native

Odoo API

Firestore



WhatsApp Smart Bot

بوت واتساب الذكي

Q1-Q2

- Instant auto-reply for: balance due, invoice copy, schedule
- Absence reporting via WhatsApp message
- Nafath contract signing reminder & link
- Upcoming events and exam schedule reminders
- Escalation to human agent if query unresolved in 2 min
- Arabic & English NLP understanding

WhatsApp Cloud API

NLP

Odoo



Bus Tracking — Parent View

تتبع الحافلة من منظور الولي

Q2-Q3

- Live map showing bus location, speed & ETA
- "Bus leaving school" push notification
- "Your child has boarded" RFID confirmation alert
- "Bus has arrived at your area" geo-fence alert
- Trip history & driver rating after each trip
- Emergency SOS button for drivers → school admin

GPS

RFID

Google Maps

Push Notify



Video Tutorials & Digital Onboarding

فيديوهات تعليمية وإعداد رقمي

Q3

- Arabic-language screencasts: portal login & Nafath signing
- "How to pay" step-by-step video for Amazon / Mada
- Bus tracking app tutorial for parents
- Document upload guide per student type
- Distributed via WhatsApp + embedded in parent portal
- Subtitles in English for non-Arabic-speaking families

Screen Record

YouTube Private

WhatsApp

Vision 2030 Aligned

متوافق مع رؤية المملكة 2030

Every initiative in this program directly supports the Kingdom's commitment to world-class education and digital economy.



Education Quality

جودة التعليم
Smart classrooms, AI-personalized learning, and early intervention systems raise student outcomes and align with national curriculum goals.



Sustainability

الاستدامة البيئية
IoT smart lighting and HVAC systems reduce energy consumption by up to 35%. Paperless processes save thousands of printed documents annually.



Digital Economy

الاقتصاد الرقمي
Paperless operations, e-invoicing (ZATCA Phase 2), digital payments, and cloud infrastructure support Saudi Arabia's digital transformation agenda.



National Identity & Security

الهوية الوطنية والأمن
Nafath integration enforces legal identity verification for all contracts. Biometric campus access enhances student safety in alignment with NCA standards.

100%

PAPERLESS
ENROLLMENT
تسجيل بلا ورق

Live

BUS TRACKING
تتبع الحافلات

-35%

ENERGY CONSUMPTION
استهلاك الطاقة

P2

ZATCA COMPLIANT
متوافق مع الزكاة

+90%

PARENT SATISFACTION
رضا الوالدين

2030

VISION ALIGNED
رؤية 2030

<1s

GATE ENTRY TIME
وقت دخول البوابة